# Seminar: Deep Learning for Molecular Biology

Alice McHardy, Giorgos Kallergis, Mohammad Hadi Foroughmand Araabi

Helmholtz Center for Infection Research & TU Braunschweig
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# Takeaways

Basic Knowledge of Machine Learning: Understanding of fundamental concepts in machine learning.

- A hot topic in both scientific research and industry applications.
- Wide-ranging impact across various domains, from healthcare to finance.

Basic Knowledge of Deep Learning: Familiarity with the principles of deep learning.

- Specialized models built upon neural network architectures.
- Remarkable success in tackling complex and challenging problems across domains.

Basic Knowledge of Bioinformatics

- Understanding of bioinformatics principles and applications.
- Awareness of how machine learning and deep learning are applied in bioinformatics research.

Experience in Problem Solving:

- Practical experience in applying machine learning and deep learning techniques to bioinformatics problems.

# Seminar overview

Max. number of participants: 10

Language: English

Designated for Bachelor and Master students of Computer Science and Data Science

Deliverables:

- Joint presentation of the chosen topic (50% by each student, on seminar day)
  - covering practical (implementation in Python) and theoretical parts (split topics).
  - Presentation of 40 minutes: 20 mins theory, 20 mins results plus showing the codes

# Seminar overview

- 7-10 page scientific report
  - written jointly, 50% by each student
  - Introduction, methods, results, discussion, literature (scientific references), acknowledgements (in report and table)
  - Introduction should cover both biology (superficially) and deep learning theory
  - Acknowledgement of the used tools (which and how, e.g. ChatGPT), you should take full ownership of the report, it means understand and approve every sentence there and know every reference you cite.
- Attend midterm meeting plus at least one meeting with the assistants
- Due Dates
  - First report version: 1 week before midterm meeting
  - Final report version: 2 weeks before the seminar date
- Grade= 50% presentation, 50% report

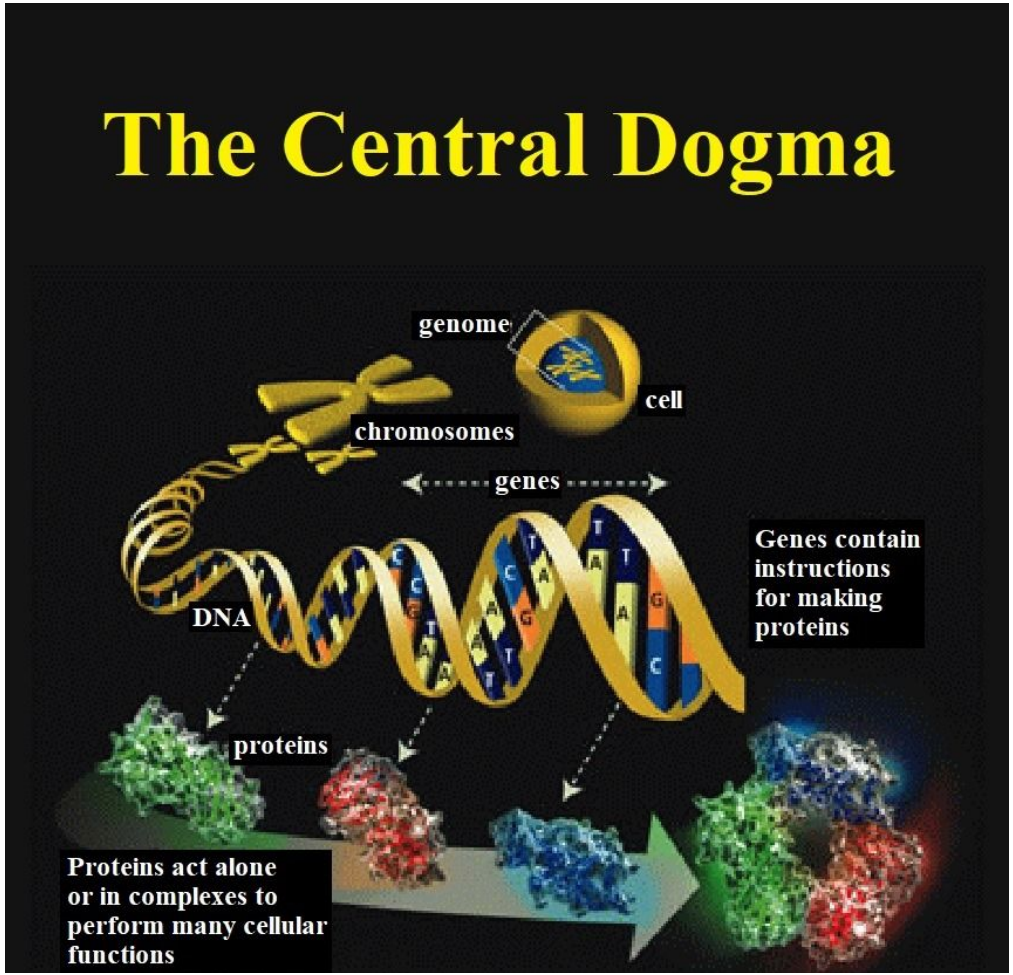
# Introduction to Bioinformatics, Machine Learning, and Deep Learning

# What is bioinformatics

(Current) Biology =

- DNA + RNA + Protein + Interactions

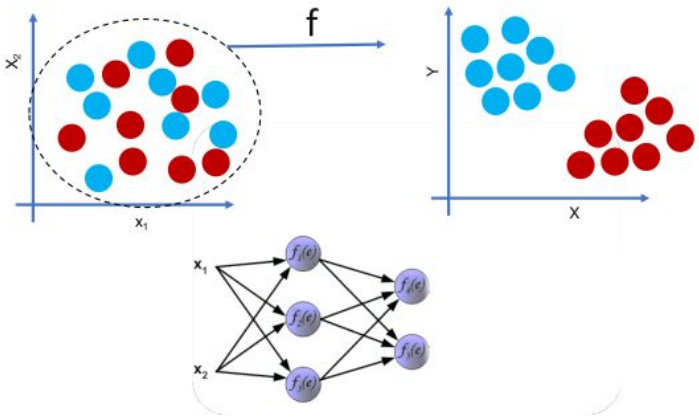
Bioinformatics: Computational analysis of the biological data



# Machine Learning & Representation learning

**Machine Learning** is a subfield of artificial intelligence focused on developing algorithms that enables systems to learn patterns, representations, or behaviors from empirical data

- Constructs models that generalize from training data to unseen instances
- Includes supervised, unsupervised, and reinforcement learning

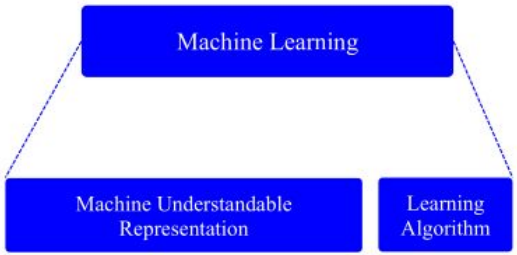



**Representation learning**:

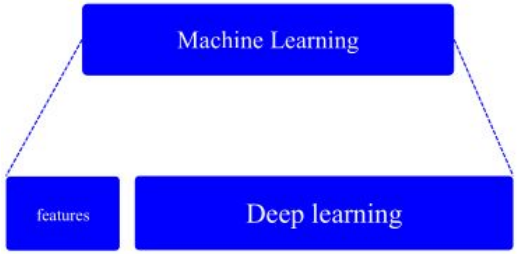
- Data can be represented as points in n-dimensional space (of features)
- Learns hierarchical or latent representations that capture underlying structures in data
  - Then, we can solve several problems, e.g. classification.

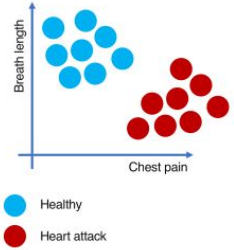
# What is machine learning

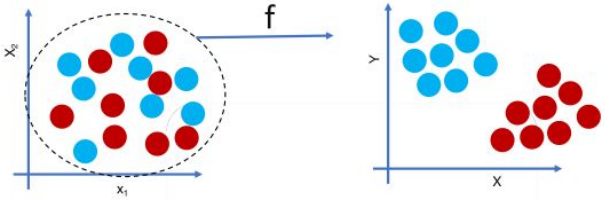
- We alter data appearance to be interpretable by the audience.
  - Machine as the audience? Numerical values, vectors, matrices
- Finding a proper representation has been critical in machine learning

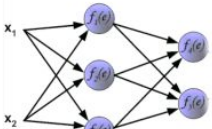



Domain knowledge
for designing representation



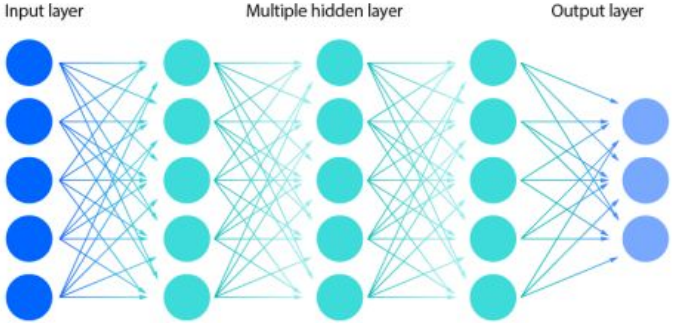



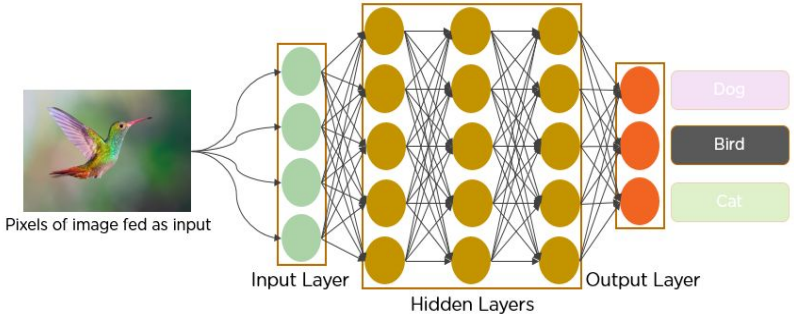



# Specific ML models: NN, RNN, CNN

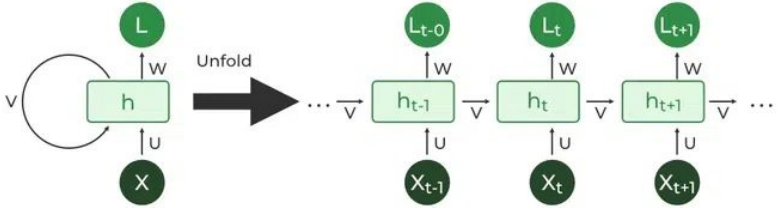
## Neural network (NN): Inspired by brain




## Convolutional Neural Network (CNN): NN with fewer (redundant) parameters

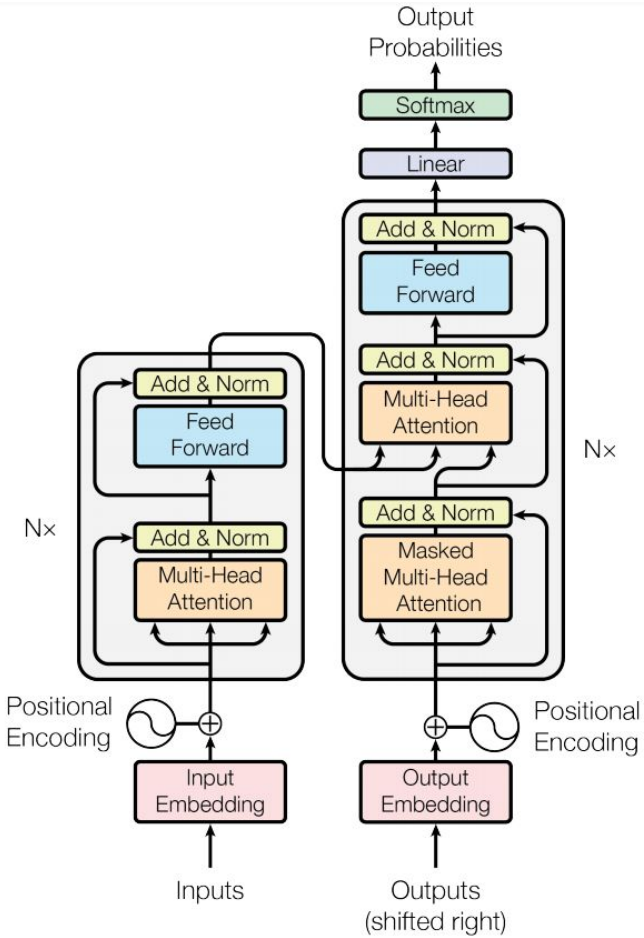



## Recurrent Neural Network (RNN): Dealing with sequential information



# What is a transformer?

- A specific architecture with breakthrough performances
- Encoder/Decoder architecture
- Used in a wide range of applications
- Effective
- Highly parallelizable
- Ideal for transfer learning




Transformers model architecture.

Vaswani et al, Attention is all you need. (NIPS 2017)

# Applications of deep learning

Computer Vision

Natural Language Processing

Bioinformatics

Advertising

...

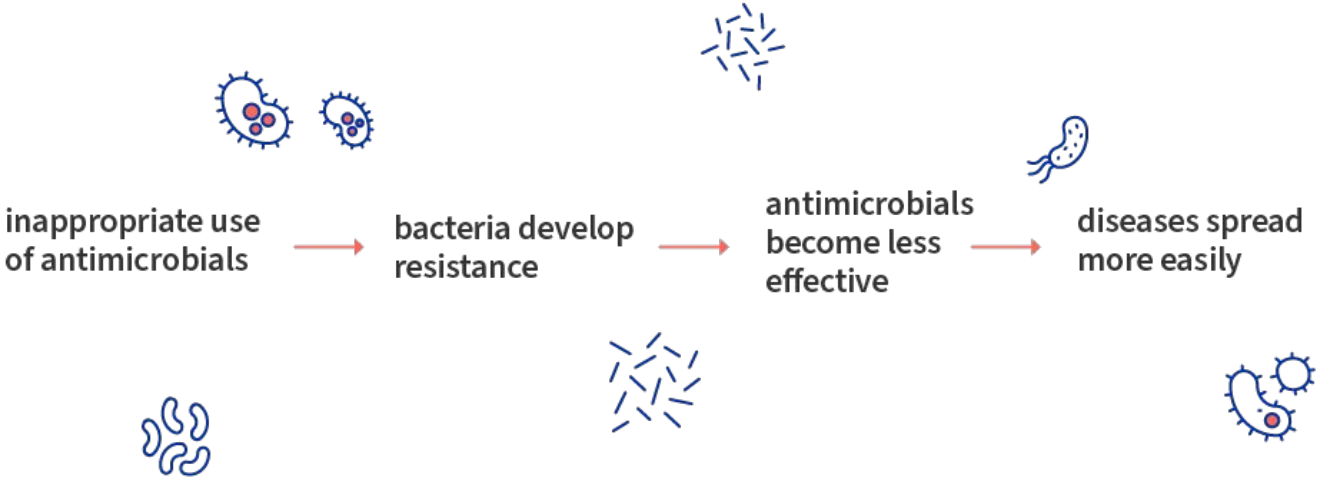
# Your task

- Find a partner to form a group
- Choose a topic (from provided list)
- Study the topic
  - Some literature is provided
  - Meet and consult with the lecturers (at least once)
  - Implement and test your topic with the dataset (if applicable)
  - Evaluate (Metrics, precision/recall, TP, TN, ...)
- Create a written report
  - **Using AI tools is permitted; please detail their use in the acknowledgements section!**
- Present your topic on the presentation day

# Implementation Tasks

## Antimicrobial Resistance of pathogens (AMR)

"AMR occurs when bacteria, viruses, fungi and parasites change over time and no longer respond to medicines making infections harder to treat and increasing the risk of disease spread, severe illness and death." WHO



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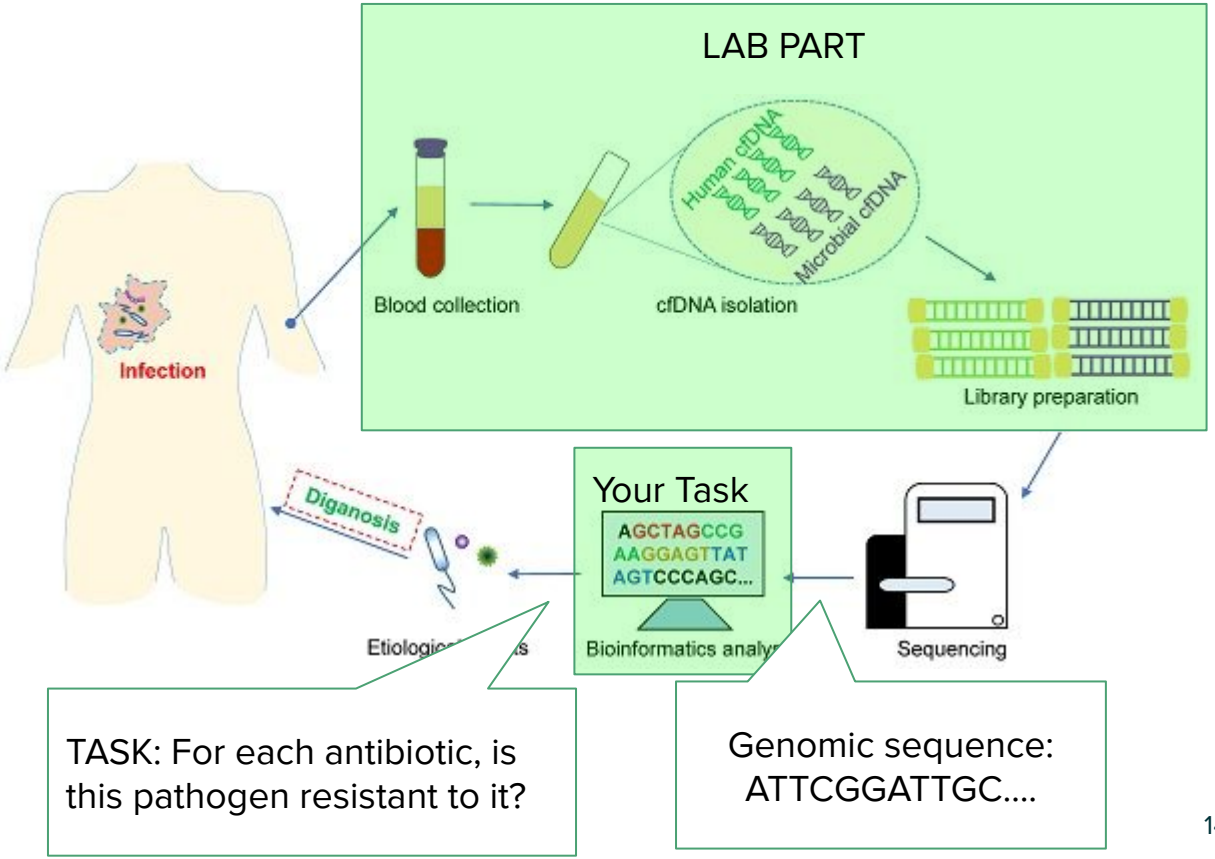


Source: https://www.thno.org/v10p5501.htm

# Datasets description (First dataset)

- Git repository: https://github.com/hzi-bifo/seminar-dlmb-2026-summer-public
- `Staphylococcus_aureus_cefoxitin_pbp4` dataset
  - Predict AMR for a pathogen (*Staphylococcus aureus*) against an antibiotic (Cefoxitin) given the genomic sequence of one gene (pbp4)
    - Note that each pathogen's genome is given as a set of contigs
  - Training set (800 samples) and test set (99 samples)
  - Labels: 0 (non-resistant), 1 (resistant)
- This dataset is the mandatory dataset.

# Datasets description (Second dataset)

- `Klebsiella_pneumoniae_aztreonam_gene_bla` dataset
  - Predict AMR for a pathogen (*Klebsiella_pneumoniae*) against an antibiotic (Aztreonam) given the genomic sequence of a few of its genes as input
    - bla_1, bla_2, bla_3, ampC, ampD, ampE, ampG, ampR, acrB, acrA, acrR, baeR, baeS, cpxA, cpxR, ramA, cpxA, cpxR
  - Training set (1136 samples) and test set (142 samples)
  - Labels: 0 (non-resistant), 1 (resistant)
- This dataset is a mandatory dataset.

# Datasets description (Optional dataset)

- `Klebsiella_pneumoniae_aztreonam` dataset
  - Predict AMR for a pathogen (*Klebsiella pneumoniae*) against an antibiotic (Aztreonam) given the genomic sequence of all its genes as input
  - Training set (1136 samples) and test set (142 samples)
  - Labels: 0 (non-resistant), 1 (resistant)
- This dataset is an optional dataset.

# Topics

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- Feed-forward Neural Networks and back propagation [2 students]
- Convolutional Neural Networks (CNNs) [2 students]
- Recurrent Neural Networks and LSTMs [2 students]
- Transformers - Encoder [2 students]
- Transformers - Decoder, Encoder-Decoder [2 students]

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Goals: Getting familiar with the basics of neural networks

- Introduction to linear classification, multilayer perceptron (MLP), and back propagation algorithm
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- Linear prediction: Lec. 1,2 at (https://bit.ly/1DIpc51) and (Chapter 4: https://stanford.io/2voWjra)
- G. Hinton's lecture 2, 3: https://bit.ly/3TNBPqw
- Lecture from U of Waterloo: https://bit.ly/2A2mzgN
- Stanford Tutorial: https://stanford.io/1FRrkZw
- Practicals: In Keras (keras.io) or Pytorch (pytorch.org)

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- Stanford Tutorial: https://stanford.io/1FRrkZw
- A more advanced reference: deeplearningbook.org
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- Representation learning: https://arxiv.org/pdf/1206.5538.pdf
- Transformer paper: https://arxiv.org/abs/1706.03762
- Simple explanation of transformers: https://jalammar.github.io/illustrated-transformer/
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Any questions? Contact us:

- mohammad-hadi.foroughmand-araabi@helmholtz-hzi.de
- georgios.kallergis@helmholtz-hzi.de

Thank you for your attention
Good luck!

# References


- Deep learning in Life Science Youtube series
- AMR image https://www.consilium.europa.eu/en/infographics/antimicrobial-resistance/